

Skill_Category	Skill	Importance_Level	Used_By_Companies	Relevant_Domains	Prerequisite_Skills	Portfolio_Project	Interview_Relevance	Learning_Path
Programming Language	Python	Essential	Google, Microsoft, OpenAI, Amazon, Meta, NVIDIA	AI, ML, Data Engineering, Backend	Programming Basics	Build AI chatbot, data pipeline, automation tools	Very High	Python syntax -> OOP -> Libraries -> Production APIs
Programming Language	Java	Essential	Amazon, Google, Microsoft, Salesforce, Oracle	Backend, Enterprise Software, Cloud	Programming Basics	Build scalable REST API system	Very High	Java Core -> Collections -> Spring Boot -> Distributed Systems
Programming Language	C++	Essential	NVIDIA, Google, Apple, Tesla, Meta, Intel	Systems, AI Infrastructure, Embedded, Gaming	Programming Basics	Build database engine or high-performance application	Very High	C++ Basics -> Memory -> STL -> Systems Programming
Programming Language	JavaScript	Essential	Google, Meta, Amazon, Netflix, Shopify	Frontend, Full Stack, Web Applications	Programming Basics	Build full-stack web application	High	JS Basics -> Browser -> Frameworks
Programming Language	TypeScript	Important	Microsoft, Google, Airbnb, Atlassian	Frontend, Full Stack, Enterprise Apps	JavaScript	Build enterprise dashboard	High	Type System -> React -> Backend Integration
Programming Language	Go (Golang)	Important	Google, Cloudflare, Uber, Kubernetes Ecosystem	Cloud, Infrastructure, Backend	Programming Basics	Build distributed service	Very High	Go Basics -> Concurrency -> Networking
Programming Language	Rust	Emerging	Cloudflare, Microsoft, Amazon, Mozilla	Systems, Security, Infrastructure	C++ Knowledge Helpful	Build secure systems tool	High	Ownership -> Memory Safety -> Systems
Programming Language	Swift	Important	Apple	iOS Development, Consumer Apps	Programming Basics	Build iOS application	High	Swift -> iOS SDK -> SwiftUI
Programming Language	Kotlin	Important	Google, Android Companies	Mobile Development, Backend	Java Knowledge	Build Android application	High	Kotlin -> Android SDK -> Architecture
Programming Language	Ruby	Important	Shopify, GitHub, Stripe	Web Platforms, SaaS	Programming Basics	Build web application	Medium	Ruby -> Rails -> APIs
Computer Science	Data Structures and Algorithms	Essential	All Top Companies	Software Engineering	Programming Basics	Solve algorithm problems	Extreme	Arrays -> Trees -> Graphs -> Dynamic Programming
Computer Science	Object Oriented Programming	Essential	All Enterprise Companies	Backend, Software Engineering	Programming Basics	Design scalable software modules	Very High	Classes -> Interfaces -> Design Patterns
Computer Science	System Design	Essential	Google, Amazon, Netflix, Uber, Meta	Backend, Cloud, Distributed Systems	Backend Programming	Design YouTube, Uber, Netflix systems	Extreme	Components -> Scaling -> Reliability
Computer Science	Operating Systems	Essential	Google, Apple, Microsoft, NVIDIA	Systems, Cloud, Embedded	Programming Basics	Build mini operating system concepts	High	Processes -> Threads -> Memory -> Scheduling
Computer Science	Computer Networks	Essential	Google, Cloudflare, Cisco, Amazon	Cloud, Security, Networking	Programming Basics	Build network service	High	TCP/IP -> HTTP -> DNS -> Distributed Networks
Backend Development	REST APIs	Essential	Almost All Software Companies	Backend, SaaS, Mobile	Programming Language	Create production API service	Very High	HTTP -> API Design -> Authentication
Backend Development	GraphQL	Important	Meta, Shopify, Netflix	Frontend Backend Integration	REST APIs	Build GraphQL service	Medium	Schema -> Queries -> APIs
Backend Development	Microservices	Essential	Amazon, Netflix, Uber, Google	Cloud, Enterprise Software	Backend Programming	Build microservice architecture	Very High	Services -> Communication -> Deployment
Backend Development	Message Queues	Important	Uber, Amazon, LinkedIn	Distributed Systems	Backend Fundamentals	Build event-driven application	High	Kafka -> RabbitMQ -> Async Processing
Backend Development	Caching Systems	Important	Amazon, Netflix, Google	High Performance Systems	Backend Fundamentals	Build caching layer	High	Redis -> CDN -> Cache Strategies
Backend Development	Authentication and Security	Essential	All Companies	Web Security, SaaS	Backend Basics	Build secure login system	High	OAuth -> JWT -> Identity Management
Backend Development	Performance Optimization	Important	Google, Meta, Netflix	Large Scale Systems	Programming Fundamentals	Optimize high traffic application	High	Profiling -> Algorithms -> Architecture
Database	SQL	Essential	All Companies	Backend, Data, Analytics	Programming Basics	Build analytics database system	Extreme	Queries -> Joins -> Indexing
Database	PostgreSQL	Important	Stripe, GitHub, Many SaaS Companies	Backend Applications	SQL	Build production backend database	High	Schema -> Optimization -> Transactions
Database	MySQL	Important	Oracle, Amazon, Enterprise Companies	Backend Systems	SQL	Build transactional application	High	Queries -> Storage Engine
Database	NoSQL Databases	Important	Amazon, Google, Meta	Distributed Systems	Database Basics	Build scalable storage system	High	MongoDB -> DynamoDB -> Cassandra
Database	Database Internals	Advanced	Google, MongoDB, Oracle	Database Engineering	SQL and Systems	Build mini database engine	Very High	Storage -> Indexing -> Query Processing
Skill_Category	Skill	Importance_Level	Used_By_Companies	Relevant_Domains	Prerequisite_Skills	Portfolio_Project	Interview_Relevance	Learning_Path
AI Fundamentals	Machine Learning Basics	Essential	Google, Microsoft, OpenAI, Meta, NVIDIA, Amazon	AI, Data Science	Python, Mathematics, Statistics	Build prediction model	Very High	Python -> Statistics -> ML Algorithms
AI Fundamentals	Deep Learning	Essential	OpenAI, Google DeepMind, NVIDIA, Meta	AI Research, Computer Vision, NLP	Python, Linear Algebra, ML Basics	Train neural network model	Very High	Neural Networks -> CNN -> Transformers
AI Fundamentals	Mathematics for AI	Essential	OpenAI, NVIDIA, Google	AI Research	Mathematics Basics	Implement ML algorithms from scratch	High	Linear Algebra -> Probability -> Calculus
AI Fundamentals	Statistics	Essential	Amazon, Uber, Netflix, Airbnb	Data Science, Analytics, ML	Mathematics	A/B testing analysis project	High	Probability -> Distribution -> Hypothesis Testing
Machine Learning	Supervised Learning	Essential	Google, Microsoft, Amazon	AI Applications	ML Basics	Classification and regression projects	High	Regression -> Trees -> Ensemble Models
Machine Learning	Unsupervised Learning	Important	Meta, Netflix, Spotify	Recommendations, Analytics	ML Basics	Customer segmentation system	Medium	Clustering -> Dimensionality Reduction
Machine Learning	Reinforcement Learning	Advanced	DeepMind, OpenAI, Tesla	Robotics, Autonomous Systems	Deep Learning	Train game-playing agent	High	MDPs -> Q Learning -> Policy Optimization
Machine Learning	Feature Engineering	Important	Amazon, Uber, Financial Companies	Applied ML	Statistics	Build optimized ML pipeline	High	Data Processing -> Feature Selection
Machine Learning	Model Evaluation	Essential	All AI Companies	AI Engineering	ML Basics	Compare ML models	High	Metrics -> Validation -> Testing
Deep Learning	PyTorch	Essential	OpenAI, Meta, NVIDIA	AI Research, LLMs	Python Deep Learning	Train neural network model	Very High	Tensors -> Models -> Training
Deep Learning	TensorFlow	Important	Google, Enterprise AI Teams	AI Applications	Python ML	Deploy ML application	High	Graphs -> Training -> Deployment
Deep Learning	CUDA	Advanced	NVIDIA, AI Infrastructure Companies	GPU Computing	C++, Python	GPU accelerated application	Very High	GPU Architecture -> Kernels -> Optimization
Deep Learning	Neural Network Architecture	Essential	Google, OpenAI, Meta	AI Research	Deep Learning	Design custom neural network	High	CNN -> RNN -> Transformers
Generative AI	Large Language Models (LLMs)	Essential	OpenAI, Anthropic, Google, Microsoft, Meta	Generative AI	Deep Learning NLP	Build LLM application	Extreme	Transformers -> Fine Tuning -> Deployment
Generative AI	Prompt Engineering	Important	OpenAI, Microsoft, Enterprise AI Companies	AI Applications	LLM Basics	Create AI workflow automation	Medium	Prompt Design -> Evaluation
Generative AI	RAG (Retrieval Augmented Generation)	Essential	OpenAI, Microsoft, Databricks	Enterprise AI	LLMs, Databases	Build document search AI assistant	Very High	Embeddings -> Vector DB -> Retrieval
Generative AI	Vector Databases	Important	OpenAI Ecosystem, Databricks, AI Startups	AI Search	Database Basics	Semantic search engine	High	Embeddings -> Indexing -> Querying
Generative AI	LLM Fine Tuning	Advanced	OpenAI, Anthropic, Meta	AI Models	Deep Learning	Fine tune open-source model	Very High	Datasets -> Training -> Evaluation
Generative AI	AI Agents	Emerging	OpenAI, Anthropic, Microsoft	Automation Systems	LLMs, APIs	Build autonomous AI agent	Very High	Tools -> Planning -> Memory
Natural Language Processing	NLP Fundamentals	Essential	Google, Meta, OpenAI	Language AI	Python ML	Text classification system	High	Tokenization -> Embeddings -> Models
Natural Language Processing	Transformers	Essential	OpenAI, Google, Anthropic	LLMs	Deep Learning	Implement transformer model	Extreme	Attention -> Encoder Decoder
Natural Language Processing	BERT Models	Important	Google, Search Companies	Search, NLP	Transformers	Build document classifier	High	Fine Tuning -> Evaluation
Computer Vision	OpenCV	Important	Tesla, Apple, NVIDIA	Vision Systems	Python Image Processing	Image recognition system	High	Images -> Features -> Detection
Computer Vision	Object Detection	Advanced	Tesla, NVIDIA, Robotics Companies	Autonomous Systems	Deep Learning	Real-time object detector	Very High	YOLO -> CNN -> Vision Transformers
Computer Vision	Image Segmentation	Advanced	Healthcare AI, Robotics	Medical Imaging, Robotics	Deep Learning	Medical image analysis system	High	Segmentation Networks
Computer Vision	3D Vision	Advanced	Apple, Meta, Robotics Companies	AR/VR, Autonomous Systems	Computer Vision	3D mapping application	High	Geometry -> Sensors -> Reconstruction
ML Ops	Machine Learning Pipelines	Essential	Google, Amazon, Databricks	Production AI	ML Basics, Cloud	Deploy ML pipeline	Very High	Data -> Training -> Deployment
ML Ops	Model Deployment	Essential	Microsoft, NVIDIA, Google	AI Engineering	ML Models	Deploy prediction API	High	Docker -> APIs -> Cloud
ML Ops	ML Monitoring	Important	Enterprise AI Companies	AI Operations	Cloud Basics	Monitor production model	High	Metrics -> Drift Detection
ML Ops	MLflow	Important	Databricks Ecosystem	ML Lifecycle	Python ML	Track experiments	Medium	Experiments -> Registry -> Deployment
Data Science	Pandas	Essential	Amazon, Netflix, Uber	Analytics, ML	Python	Data analysis project	High	Data Cleaning -> Analysis
Data Science	NumPy	Essential	All AI Teams	Scientific Computing	Python	Numerical computing project	Medium	Arrays -> Vectorization
Data Science	Apache Spark	Essential	Databricks, Amazon, Netflix	Big Data Processing	Python, SQL	Large-scale data pipeline	Very High	RDD -> DataFrames -> Optimization
Data Science	Data Visualization	Important	Analytics Teams	Business Intelligence	Statistics	Business dashboard	Medium	Charts -> Storytelling
Data Science	A/B Testing	Important	Google, Netflix, Amazon	Product Analytics	Statistics	Experiment analysis	High	Experiment Design -> Metrics
Data Engineering	Data Pipelines	Essential	Amazon, Uber, Netflix	Data Platforms	SQL Python	Build ETL pipeline	Very High	Extract -> Transform -> Load
Data Engineering	Apache Kafka	Essential	LinkedIn, Uber, Confluent	Streaming Systems	Backend Basics	Real-time event system	Very High	Messaging -> Streams -> Scaling
Data Engineering	Apache Airflow	Important	Data Companies	Workflow Automation	Python	Automated data workflows	High	DAGs -> Scheduling
Skill_Category	Skill	Importance_Level	Used_By_Companies	Relevant_Domains	Prerequisite_Skills	Portfolio_Project	Interview_Relevance	Learning_Path
Cloud Computing	AWS	Essential	Amazon, Netflix, Airbnb, Stripe, Adobe	Cloud, Backend, DevOps	Linux, Networking Basics	Deploy scalable cloud application	Very High	EC2 -> S3 -> IAM -> VPC -> Advanced Services
Cloud Computing	Microsoft Azure	Essential	Microsoft, Enterprise Companies	Cloud, Enterprise Software	Networking, Cloud Basics	Deploy enterprise cloud solution	High	Azure Compute -> Storage -> Security
Cloud Computing	Google Cloud Platform (GCP)	Essential	Google, Data Companies	AI, Data Engineering, Cloud	Cloud Fundamentals	Deploy ML application	High	Compute -> Kubernetes -> AI Services
Cloud Computing	Cloud Architecture	Essential	Amazon, Google, Microsoft	Cloud Engineering, Cloud	Networking, System Design	Design multi-region application	Extreme	Availability -> Scaling -> Cost Optimization
Cloud Computing	Serverless Computing	Important	AWS, Google, Microsoft	Cloud Applications	Cloud Basics	Build serverless API	Medium	Functions -> Events -> Scaling
Cloud Computing	Infrastructure as Code	Essential	Cloud Companies	DevOps, Platform Engineering	Cloud Basics	Automated infrastructure deployment	High	Terraform -> CloudFormation
Cloud Computing	Terraform	Essential	HashiCorp, AWS, Google Cloud Users	Infrastructure Automation	Cloud Basics	Create cloud infrastructure templates	Very High	Providers -> Modules -> State Management
Cloud Computing	Cloud Networking	Essential	AWS, Google, Microsoft	Cloud Infrastructure	Networking	Design secure cloud network	High	VPC -> Subnets -> Routing
Cloud Computing	Cloud Security	Essential	Microsoft, Palo Alto, AWS	Security Engineering	Cloud Basics	Secure cloud environment	Very High	IAM -> Encryption -> Monitoring
DevOps	Linux Administration	Essential	Google, Amazon, Cloudflare, Red Hat	Infrastructure, Cloud	Operating Systems	Manage Linux server	Extreme	Shell -> Processes -> Networking
DevOps	Bash/Shell Scripting	Essential	All Infrastructure Teams	Automation	Linux Basics	Automate deployment scripts	High	Commands -> Scripts -> Automation
DevOps	Git	Essential	All Software Companies	Software Development	Programming Basics	Collaborative software project	Very High	Version Control -> Branching -> Workflows
DevOps	CI/CD Pipelines	Essential	All Technology Companies	Software Delivery	Git Basics	Automated deployment pipeline	Very High	Build -> Test -> Deploy
DevOps	Jenkins	Important	Enterprise Companies	CI/CD	Linux Git	Create build pipeline	High	Jobs -> Pipelines -> Automation

DevOps	GitHub Actions	Important	GitHub Ecosystem	CI/CD	Git Basics	Automated project deployment	High	Workflows -> Actions
DevOps	Docker	Essential	Almost All Cloud Companies	Containers, Cloud	Linux Basics	Containerize application	Extreme	Images -> Containers -> Networks
DevOps	Kubernetes	Essential	Google, Microsoft, Amazon, Cloud Native Companies	Container Orchestration	Docker, Networking	Deploy production cluster	Extreme	Pods -> Services -> Deployments
DevOps	Helm	Important	Kubernetes Teams	Cloud Native	Kubernetes	Package Kubernetes applications	Medium	Charts -> Releases
DevOps	Service Mesh	Advanced	Google, Uber, Enterprise Cloud	Microservices	Kubernetes	Build secure microservice network	High	Istio -> Traffic Management
Infrastructure	Site Reliability Engineering (SRE)	Essential	Google, Netflix, Amazon	Large Scale Systems	Cloud, Linux, Programming	Build reliable production service	Very High	Monitoring -> Reliability -> Automation
Infrastructure	Monitoring Systems	Essential	Datadog, Google, Netflix	Operations	Cloud Basics	Create monitoring dashboard	High	Metrics -> Alerts -> Observability
Infrastructure	Prometheus	Important	Cloud Native Companies	Monitoring	Kubernetes	Monitor Kubernetes cluster	High	Metrics -> Queries -> Alerts
Infrastructure	Grafana	Important	Cloud Companies	Visualization	Monitoring Basics	Build operations dashboard	Medium	Dashboards -> Visualization
Infrastructure	Logging Systems	Essential	Uber, Amazon, Netflix	Distributed Systems	Backend Basics	Centralized logging platform	High	Logs -> Search -> Analysis
Infrastructure	Distributed Tracing	Advanced	Google, Uber, Cloud Companies	Microservices	Backend	Trace distributed requests	High	OpenTelemetry -> Analysis
Infrastructure	System Performance Optimization	Advanced	Google, Meta, NVIDIA	Systems Engineering	Operating Systems	Optimize high-performance service	Very High	Profiling -> Benchmarking
Cybersecurity	Security Fundamentals	Essential	Microsoft, Palo Alto, CrowdStrike	Cybersecurity	Networking Basics	Security assessment project	High	Threats -> Defense -> Controls
Cybersecurity	Network Security	Essential	Cisco, Cloudflare, Palo Alto	Security Infrastructure	Networking	Build secure network	Very High	Firewalls -> IDS -> VPN
Cybersecurity	Application Security	Essential	All Software Companies	Software Security	Programming	Secure web application	Very High	OWASP -> Testing -> Secure Coding
Cybersecurity	Ethical Hacking	Important	Security Companies	Cybersecurity	Networking Linux	Security lab projects	High	Recon -> Testing -> Reporting
Cybersecurity	Encryption and Cryptography	Essential	Google, Apple, Banking Companies	Security, Privacy	Mathematics	Implement encryption algorithms	High	Hashing -> Keys -> Protocols
Cybersecurity	Identity and Access Management (IAM)	Essential	AWS, Microsoft, Okta	Cloud Security	Security Basics	Build authentication system	High	Users -> Roles -> Permissions
Cybersecurity	Threat Detection	Advanced	CrowdStrike, Microsoft	Security AI	Machine Learning Security	Threat detection model	Very High	Signals -> Detection -> Response
Networking	TCP/IP	Essential	Cisco, Google, Cloudflare	Networking	Computer Networks	Build network application	Extreme	Protocols -> Packets -> Routing
Networking	HTTP/HTTPS	Essential	All Web Companies	Web Development	Networking Basics	Build web server	Very High	Requests -> Responses -> Security
Networking	DNS	Important	Cloudflare, Google	Internet Infrastructure	Networking	Build DNS resolver	High	Records -> Resolution
Networking	Load Balancing	Essential	Amazon, Netflix, Google	Cloud Systems	System Design	Build traffic distribution system	High	Algorithms -> Scaling
Networking	CDN Technology	Important	Cloudflare, Akamai, Netflix	Internet Performance	Networking	Build content delivery system	High	Caching -> Edge Computing
Networking	Network Programming	Advanced	Cisco, Cloudflare	Systems Engineering	C/C++ Networking	Build network protocol tool	High	Sockets -> Protocols
Cloud Computing	AWS	Essential	Amazon, Netflix, Airbnb, Stripe, Adobe	Cloud, Backend, DevOps	Linux, Networking Basics	Deploy scalable cloud application	Very High	EC2 -> S3 -> IAM -> VPC -> Advanced Services
Cloud Computing	Microsoft Azure	Essential	Microsoft, Enterprise Companies	Cloud, Enterprise Software	Networking, Cloud Basics	Deploy enterprise cloud solution	High	Azure Compute -> Storage -> Security
Cloud Computing	Google Cloud Platform (GCP)	Essential	Google, Data Companies	AI, Data Engineering, Cloud	Cloud Fundamentals	Deploy ML application	High	Compute -> Kubernetes -> AI Services
Cloud Computing	Cloud Architecture	Essential	Amazon, Google, Microsoft	Cloud Engineering	Networking, System Design	Design multi-region application	Extreme	Availability -> Scaling -> Cost Optimization
Cloud Computing	Serverless Computing	Important	AWS, Google, Microsoft	Cloud Applications	Cloud Basics	Build serverless API	Medium	Functions -> Events -> Scaling
Cloud Computing	Infrastructure as Code	Essential	Cloud Companies	DevOps, Platform Engineering	Cloud Basics	Automated infrastructure deployment	High	Terraform -> CloudFormation
Cloud Computing	Terraform	Essential	HashiCorp, AWS, Google Cloud Users	Infrastructure Automation	Cloud Basics	Create cloud infrastructure templates	Very High	Providers -> Modules -> State Management
Cloud Computing	Cloud Networking	Essential	AWS, Google, Microsoft	Cloud Infrastructure	Networking	Design secure cloud network	High	VPC -> Subnets -> Routing
Cloud Computing	Cloud Security	Essential	Microsoft, Palo Alto, AWS	Security Engineering	Cloud Basics	Secure cloud environment	Very High	IAM -> Encryption -> Monitoring
DevOps	Linux Administration	Essential	Google, Amazon, Cloudflare, Red Hat	Infrastructure, Cloud	Operating Systems	Manage Linux server	Extreme	Shell -> Processes -> Networking
DevOps	Bash/Shell Scripting	Essential	All Infrastructure Teams	Automation	Linux Basics	Automate deployment scripts	High	Commands -> Scripts -> Automation
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DevOps	Helm	Important	Kubernetes Teams	Cloud Native	Kubernetes	Package Kubernetes applications	Medium	Charts -> Releases
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Infrastructure	Monitoring Systems	Essential	Datadog, Google, Netflix	Operations	Cloud Basics	Create monitoring dashboard	High	Metrics -> Alerts -> Observability
Infrastructure	Prometheus	Important	Cloud Native Companies	Monitoring	Kubernetes	Monitor Kubernetes cluster	High	Metrics -> Queries -> Alerts
Infrastructure	Grafana	Important	Cloud Companies	Visualization	Monitoring Basics	Build operations dashboard	Medium	Dashboards -> Visualization
Infrastructure	Logging Systems	Essential	Uber, Amazon, Netflix	Distributed Systems	Backend Basics	Centralized logging platform	High	Logs -> Search -> Analysis
Infrastructure	Distributed Tracing	Advanced	Google, Uber, Cloud Companies	Microservices	Backend	Trace distributed requests	High	OpenTelemetry -> Analysis
Infrastructure	System Performance Optimization	Advanced	Google, Meta, NVIDIA	Systems Engineering	Operating Systems	Optimize high-performance service	Very High	Profiling -> Benchmarking
Cybersecurity	Security Fundamentals	Essential	Microsoft, Palo Alto, CrowdStrike	Cybersecurity	Networking Basics	Security assessment project	High	Threats -> Defense -> Controls
Cybersecurity	Network Security	Essential	Cisco, Cloudflare, Palo Alto	Security Infrastructure	Networking	Build secure network	Very High	Firewalls -> IDS -> VPN
Cybersecurity	Application Security	Essential	All Software Companies	Software Security	Programming	Secure web application	Very High	OWASP -> Testing -> Secure Coding
Cybersecurity	Ethical Hacking	Important	Security Companies	Cybersecurity	Networking Linux	Security lab projects	High	Recon -> Testing -> Reporting
Cybersecurity	Encryption and Cryptography	Essential	Google, Apple, Banking Companies	Security, Privacy	Mathematics	Implement encryption algorithms	High	Hashing -> Keys -> Protocols
Cybersecurity	Identity and Access Management (IAM)	Essential	AWS, Microsoft, Okta	Cloud Security	Security Basics	Build authentication system	High	Users -> Roles -> Permissions
Cybersecurity	Threat Detection	Advanced	CrowdStrike, Microsoft	Security AI	Machine Learning Security	Threat detection model	Very High	Signals -> Detection -> Response
Networking	TCP/IP	Essential	Cisco, Google, Cloudflare	Networking	Computer Networks	Build network application	Extreme	Protocols -> Packets -> Routing
Networking	HTTP/HTTPS	Essential	All Web Companies	Web Development	Networking Basics	Build web server	Very High	Requests -> Responses -> Security
Networking	DNS	Important	Cloudflare, Google	Internet Infrastructure	Networking	Build DNS resolver	High	Records -> Resolution
Networking	Load Balancing	Essential	Amazon, Netflix, Google	Cloud Systems	System Design	Build traffic distribution system	High	Algorithms -> Scaling
Networking	CDN Technology	Important	Cloudflare, Akamai, Netflix	Internet Performance	Networking	Build content delivery system	High	Caching -> Edge Computing
Networking	Network Programming	Advanced	Cisco, Cloudflare	Systems Engineering	C/C++ Networking	Build network protocol tool	High	Sockets -> Protocols
Frontend Development	HTML	Critical	Google, Microsoft, Meta, Amazon, Shopify	Web Development	Basic Programming	Build responsive website	Medium	HTML Structure -> Accessibility -> SEO
Frontend Development	CSS	Essential	Google, Meta, Airbnb, Shopify	Frontend Engineering	HTML	Build modern UI design	Medium	Layouts -> Flexbox -> Grid -> Animations
Frontend Development	JavaScript	Essential	Google, Meta, Netflix, Airbnb	Frontend, Full Stack	Programming Basics	Build interactive web application	Very High	JS Core -> Browser APIs -> Frameworks
Frontend Development	React	Essential	Meta, Microsoft, Netflix, Airbnb	Frontend Engineering	JavaScript	Build production dashboard	Very High	Components -> State -> Hooks
Frontend Development	Angular	Important	Google, Enterprise Companies	Enterprise Frontend	JavaScript	Build enterprise web app	High	Components -> Services -> Architecture
Frontend Development	Vue.js	Important	Startups, SaaS Companies	Frontend Applications	JavaScript	Build SPA application	Medium, Components -> State Management	
Frontend Development	Next.js	Important	Vercel, Modern Web Companies	Full Stack Web	React	Build full-stack application	High	SSR -> Routing -> APIs
Frontend Development	Web Performance Optimization	Advanced	Google, Meta	Frontend Infrastructure	JavaScript	Optimize website speed	High	Rendering -> Caching -> Optimization
Frontend Development	UI/UX Principles	Important	Apple, Google, Adobe	Product Engineering	Frontend Basics	Design user-friendly application	Medium	Design Systems -> Accessibility
Mobile Development	Android Development	Essential	Google, Samsung, Uber	Mobile Applications	Java/Kotlin	Build Android app	High	Android SDK -> Architecture
Mobile Development	iOS Development	Essential	Apple, Uber, Airbnb	Mobile Applications	Swift	Build iOS app	High	Swift -> UIKit -> SwiftUI
Mobile Development	Flutter	Important	Google, Startups	Cross Platform Apps	Dart Programming	Build cross-platform app	Medium	Widgets -> State Management
Mobile Development	React Native	Important	Meta, Startups	Mobile Applications	JavaScript React	Build mobile application	Medium	Components -> Native Modules
Mobile Development	Mobile Architecture	Advanced	Apple, Google	Large Mobile Apps	Mobile Development	Design scalable mobile app	High	MVC -> MVVM -> Clean Architecture
Embedded Systems	C Programming	Essential	Intel, Qualcomm, Tesla, Bosch	Embedded Systems	Programming Basics	Microcontroller project	Very High	Pointers -> Memory -> Hardware
Embedded Systems	C++ Programming	Essential	NVIDIA, Apple, Tesla	Systems, Robotics	C Basics	Embedded software project	Very High, OOP -> Performance -> Hardware	
Embedded Systems	Embedded Linux	Essential	Qualcomm, Automotive Companies	IoT, Automotive	Linux Basics	Embedded Linux device	High	Kernel -> Drivers -> Applications
Embedded Systems	Microcontrollers	Important	Bosch, Siemens, Robotics Companies	IoT Hardware	C Programming	Build sensor system	High	Arduino -> STM32 -> ARM

Embedded Systems	RTOS	Advanced	Automotive, Aerospace	Real-Time Systems	Embedded C	Build real-time application	High	Tasks -> Scheduling -> Interrupts
Embedded Systems	Device Drivers	Advanced	Intel, NVIDIA, Qualcomm	Hardware Software Integration	C Programming	Write hardware driver	Very High	Kernel -> Hardware Communication
Embedded Systems	Firmware Development	Essential	Apple, Tesla, Samsung	Hardware Products	C/C++	Create device firmware	High,Hardware -> Boot Process -> Firmware"	
Robotics	Robot Operating System (ROS)	Essential	Tesla, ABB, Robotics Companies	Robotics	C++, Python	Build autonomous robot	Very High	Nodes -> Sensors -> Control
Robotics	Robot Control Systems	Advanced	ABB, Tesla	Robotics Automation	Mathematics, C++	Robot movement controller	High	PID -> Motion Planning
Robotics	Sensors and Actuators	Essential	Robotics, Automotive	Physical Systems	Embedded Basics	Sensor-based robot	High	Sensors -> Calibration -> Integration
Robotics	SLAM (Mapping)	Advanced	Waymo, Robotics Companies	Autonomous Systems	Computer Vision Robotics	Build mapping robot	Very High	Localization -> Mapping
Robotics	Computer Vision for Robotics	Advanced	Tesla, NVIDIA	Autonomous Machines	Deep Learning Vision	Object detection robot	Very High	Vision Models -> Deployment
Robotics	Path Planning	Advanced	Tesla, Waymo	Autonomous Vehicles	Algorithms Robotics	Autonomous navigation system	High	Graph Search -> Optimization
Semiconductors	Computer Architecture	Essential	Intel, AMD, Apple, NVIDIA	Processors	Operating Systems Basics	CPU simulator project	Very High	CPU -> Memory -> Performance
Semiconductors	VLSI Design	Advanced	Intel, AMD, Qualcomm	Chip Design	Digital Electronics	Design hardware circuit	High	Logic -> RTL -> Verification
Semiconductors	Verilog/SystemVerilog	Essential	NVIDIA, Intel, AMD	Hardware Design	Digital Logic	Create processor module	Very High	RTL -> Simulation -> Testing
Semiconductors	FPGA Programming	Advanced	Intel, Xilinx, Defense Companies	Hardware Acceleration	Digital Design	FPGA accelerator project	High	HDL -> FPGA -> Optimization
Semiconductors	GPU Computing	Advanced	NVIDIA, AMD	AI Hardware	C++ Programming	CUDA accelerated application	Extreme	GPU Architecture -> Parallel Programming
Semiconductors	Parallel Computing	Advanced	NVIDIA, Google	AI Systems	Computer Architecture	Parallel processing project	High	Threads -> GPU -> Optimization
Automotive Engineering	AUTOSAR	Important	Bosch, Toyota, BMW	Automotive Software	Embedded C	Vehicle software module	High	Automotive Standards -> Software Layers
Automotive Engineering	Vehicle Communication (CAN)	Essential	Tesla, Toyota, Bosch	Automotive Systems	Embedded Systems	Vehicle communication simulator	High	CAN Protocol -> EQU Communication
Automotive Engineering	ADAS Systems	Advanced	Tesla, Waymo	Driver Assistance	Computer Vision	Driver assistance prototype	Very High	Sensors -> AI -> Control
Automotive Engineering	Battery Management Systems	Important	Tesla, Rivian	Electric Vehicles	Embedded Systems	Battery monitoring system	High	Battery -> Sensors -> Control
IoT	Internet of Things	Essential	Amazon, Siemens, Bosch	Connected Devices	Embedded Systems	Smart home device	High	Sensors -> Cloud -> Analytics
IoT	MQTT Protocol	Important	IoT Companies	Device Communication	Networking	IoT messaging system	Medium	Publish -> Subscribe
AR/VR	3D Graphics	Advanced	Apple, Meta, Sony	Spatial Computing	Mathematics Programming	3D rendering application	High	Geometry -> Rendering -> Optimization
AR/VR	Unity Engine	Important	Gaming, XR Companies	Virtual Reality	C# Programming	VR application	High	Scenes -> Physics -> Interaction
AR/VR	Unreal Engine	Important	Epic Games, Gaming Companies	Games, Metaverse	C++ Programming	3D environment project	High	Rendering -> Engine Programming
AR/VR	Spatial Computing	Emerging	Apple, Meta	AR/VR	Computer Vision	AR experience prototype	Very High	Sensors -> Vision -> Interaction
Software Architecture	Design Patterns	Essential	Google, Microsoft, Amazon, Salesforce	Software Engineering	OOP Programming	Build scalable application architecture	High	Factory -> Strategy -> Observer -> Dependency Injection
Software Architecture	Clean Architecture	Important	Enterprise Software Companies	Backend, Mobile, SaaS	OOP, System Design	Design maintainable application	High	Layers -> Dependencies -> Testing
Software Architecture	Domain Driven Design	Advanced	Amazon, Microsoft, Enterprise Companies	Large Software Systems	Software Architecture	Build business domain platform	High	Entities -> Aggregates -> Services
Software Architecture	Event Driven Architecture	Advanced	Uber, Netflix, LinkedIn	Distributed Systems	Message Queues	Build event processing system	Very High	Events -> Brokers -> Consumers
Software Architecture	API Design	Essential	Stripe, Google, Microsoft	Backend Systems	Backend Development	Create developer-friendly API	Very High	REST -> GraphQL -> Versioning
Software Architecture	Scalable System Design	Essential	Google, Amazon, Meta, Netflix	Cloud Systems	Backend Basics	Design YouTube/Uber/Netflix system	Extreme	Requirements -> Architecture -> Scaling
Software Architecture	Reliability Engineering	Essential	Google, Netflix, Amazon	Infrastructure	Cloud, SRE	Build highly available service	Very High	Failure Handling -> Monitoring -> Recovery
Software Architecture	High Availability Systems	Advanced	Cloud Companies	Distributed Systems	System Design	Multi-region application	High	Replication -> Failover -> Recovery
Software Architecture	Database Scaling	Advanced	Amazon, Google, Snowflake	Data Systems	Databases	Distributed database design	Very High	Sharding -> Replication -> Partitioning
Software Architecture	System Optimization	Advanced	NVIDIA, Google, Meta	Performance Engineering	Programming Fundamentals	Optimize high-performance system	High	Profiling -> Benchmarking -> Optimization
Software Testing	Unit Testing	Essential	All Software Companies	Software Development	Programming	Create tested application	High	Test Frameworks -> Automation
Software Testing	Integration Testing	Important	Enterprise Companies	Backend Systems	APIs	Test distributed application	High	Services -> Databases -> APIs
Software Testing	Test Automation	Important	Google, Microsoft	QA Engineering	Automation framework	Load testing project	Medium	Frameworks -> CI/CD
Software Testing	Performance Testing	Advanced	Amazon, Netflix	Large Scale Systems	System Design	Load testing project	High	Stress -> Benchmark -> Optimize
Development Tools	Git Advanced	Essential	All Technology Companies	Software Development	Git Basics	Open source contribution	Very High	Branching -> Merging -> Workflows
Development Tools	Code Review Skills	Essential	Google, Meta, Microsoft	Engineering Teams	Programming	Review open source code	High	Standards -> Feedback -> Quality
Development Tools	IDE Productivity	Important	All Developers	Software Development	Programming	Developer workflow optimization	Medium	Debugging -> Extensions -> Automation
Development Tools	Documentation Skills	Essential	All Companies	Engineering Collaboration	Technical Knowledge	Write technical documentation	High	Architecture Docs -> API Docs
Research Skills	Technical Reading	Important	OpenAI, Google Research, NVIDIA	AI Research, Engineering	Technical Fundamentals	Summarize research papers	High	Papers -> Implementation -> Analysis
Research Skills	Academic Papers	Advanced	DeepMind, OpenAI, Meta AI	AI Research	Mathematics, ML	Reproduce research paper	Very High	Read -> Implement -> Improve
Research Skills	Experiment Design	Important	AI Labs, Data Companies	AI, Product Analytics	Statistics	Run ML experiments	High	Hypothesis -> Testing -> Evaluation
Research Skills	Scientific Writing	Important	Research Organizations	AI Research	Technical Knowledge	Publish technical article	Medium	Structure -> Evidence -> Explanation
Open Source	Open Source Contribution	Important	Google, Microsoft, Red Hat	Software Engineering	Programming	Contribute to GitHub projects	High	Find Project -> Issues -> Pull Requests
Open Source	Maintaining Projects	Advanced	GitHub, Linux Community	Developer Ecosystem	Software Development	Maintain open-source library	High	Community -> Releases -> Documentation
Open Source	Reading Large Codebases	Essential	Big Tech Companies	Software Engineering	Programming	Study major repositories	Very High	Architecture -> Modules -> Dependencies
Product Engineering	Product Thinking	Important	Apple, Google, Airbnb	Product Development	Software Skills	Build user-focused product	High	Users -> Problems -> Solutions
Product Engineering	User Experience Thinking	Important	Apple, Meta, Adobe	Consumer Products	Frontend Basics	Improve application UX	Medium	Research -> Design -> Feedback
Product Engineering	A/B Testing	Important	Google, Netflix, Amazon	Product Analytics	Statistics	Experiment framework	High	Metrics -> Experiments -> Decisions
Product Engineering	Analytics Skills	Important	Uber, Airbnb, Spotify	Product Engineering	SQL	Build product analytics dashboard	Medium	Metrics -> Insights
Leadership Skills	Communication	Essential	All Top Companies	Engineering Leadership	Technical Knowledge	Technical presentation	Very High	Writing -> Speaking -> Collaboration
Leadership Skills	Technical Leadership	Advanced	Senior Engineering Roles	Software Teams	Experience	Lead engineering project	High	Planning -> Mentoring -> Execution
Leadership Skills	Mentoring	Important	Google, Microsoft	Engineering Culture	Experience	Guide junior developers	Medium	Teaching -> Feedback
Leadership Skills	Project Management	Important	Enterprise Companies	Engineering Management	Communication	Manage software project	Medium	Planning -> Delivery
Career Preparation	Resume Engineering	Essential	All Companies	Career Growth	Projects	Create technical resume	High	Impact Writing -> Metrics -> Projects
Career Preparation	LinkedIn Profile Optimization	Important	All Companies	Professional Branding	Experience	Build professional presence	Medium	Profile -> Networking -> Content
Career Preparation	Technical Interview Preparation	Essential	Google, Amazon, Meta	Software Engineering	DSA Basics	Solve interview problems	Extreme	Algorithms -> Mock Interviews
Career Preparation	Behavioral Interviews	Essential	Amazon, Microsoft, Google	All Roles	Communication	Prepare leadership examples	Very High	STAR Method -> Story Building
Career Preparation	System Design Interviews	Essential	Senior Software Roles	Backend, Cloud	Architecture Basics	Design large-scale systems	Extreme	Requirements -> Components -> Tradeoffs
Certifications	Cloud Certifications	Important	AWS, Microsoft, Google Cloud Companies	Cloud Engineering	Cloud Basics	Deploy certified cloud projects	Medium	AWS Solutions Architect -> Azure -> GCP
Certifications	Kubernetes Certification	Important	Cloud Native Companies	DevOps	Docker Kubernetes	Manage Kubernetes cluster	High	Containers -> Kubernetes -> Operations
Certifications	Cybersecurity Certifications	Important	Security Companies	Cybersecurity	Networking	Security lab projects	Medium	Security Fundamentals -> Advanced
Certifications	AI Certifications	Optional	AI Companies	Machine Learning	ML Basics	AI portfolio projects	Medium	ML Foundations -> Deep Learning
Portfolio Projects	Distributed System Project	Essential	Google, Amazon, Netflix	Backend Systems	System Design	Build URL shortener/cache/message system	Extreme	Architecture -> Implementation -> Scaling
Portfolio Projects	AI Application Project	Essential	OpenAI, Anthropic, Microsoft	AI Engineering	Python ML	Build RAG AI assistant	Very High	LLM -> Retrieval -> Deployment
Portfolio Projects	Cloud Infrastructure Project	Essential	AWS, Google, Microsoft	Cloud Engineering	Cloud Basics	Deploy production platform	Very High	Docker -> Kubernetes -> Monitoring
Portfolio Projects	Open Source Project	Important	GitHub, Red Hat, Microsoft	Software Engineering	Programming	Contribute to major repository	High	Contribution -> Review -> Release
Portfolio Projects	Mobile Application	Important	Apple, Google	Mobile Development	Mobile Skills	Build production mobile app	High	UI -> Backend -> Deployment
Portfolio Projects	Embedded Project	Important	Tesla, NVIDIA, Qualcomm	Hardware Systems	C/C++	Build IoT device	High	Hardware -> Firmware -> Cloud